Predicting Lap Times in Formula 1 : A Deep Learning Approach Using LSTM Networks

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***Abstract*— One of the most critical factors influencing success in Formula 1 racing is the ability to make strategic decisions at the optimal moment. These decisions—such as pit stop timing and tire selection—are deeply tied to lap time dynamics. In this study, we propose a Long Short-Term Memory (LSTM)-based time series model for predicting the next lap time of a driver using real-world multidimensional race data. The model is trained on data from the 2021–2023 seasons and evaluated on the 2024 season. A comprehensive data preprocessing and feature engineering pipeline is introduced, integrating race-specific variables with driver and team characteristics to enhance predictive performance. The proposed architecture effectively captures both sequential dependencies and contextual information, leading to more accurate and realistic lap time forecasts. Beyond its predictive capability, the model offers a promising foundation for real-time race assistants and decision-support systems. Future enhancements may include driver-personalized architectures, dynamic race condition modeling, and integrated strategic recommendation systems.**

Keywords*— *Formula 1, deep learning, time series, LSTM, lap time prediction

I. INTRODUCTION

Formula 1 not only represents the pinnacle of motorsports but also serves as a dynamic testing ground for real-time data analytics, engineering strategies, and artificial intelligence applications. This high-speed and highly variable environment often requires instant decisions that can ultimately determine race outcomes and championship standings. The primary objective of this study is to develop a time series model capable of predicting future lap times based on the analysis of historical Formula 1 race data. Unlike traditional approaches that focus solely on passive historical analysis, this study presents a fine-grained, AI-driven approach that evaluates each lap individually, capturing variations in driver pace, tire degradation, and track conditions over time.

Each lap in a Formula 1 race is affected by numerous dynamic factors such as tire type, pit stop timing, weather conditions, and on-track traffic. Consequently, accurately predicting driver performance is highly complex and cannot be achieved with simple linear models. Particularly during mid-race strategic decisions, the ability to estimate the upcoming lap time becomes critically important. To address this challenge, we propose a deep learning-based architecture capable of modeling sequential dependencies in the data. Long Short-Term Memory (LSTM) networks were selected for their ability to process sequential data and capture temporal patterns across laps.

The dataset used in this study consists of detailed lap-by-lap data from the 2021 to 2024 Formula 1 seasons. Through extensive preprocessing and feature engineering, new meaningful variables were derived—such as driver identity, team affiliation, tire compound, and tire age. The final model was trained in a time-series format and evaluated based on both overall and team/driver-specific performance metrics.

The main achievement of this study is the development of an LSTM-based pipeline that performs lap-level prediction with high accuracy across four Formula 1 seasons, achieving strong performance on 2024 races and demonstrating its potential as a foundation for real-time decision support.

The remainder of this paper is organized as follows: Section II reviews related work on lap time prediction and race strategy modeling. Section III describes the dataset, preprocessing, and the LSTM model architecture. Section IV presents the experimental results and analysis. Section V concludes the paper and outlines directions for future work.

II. RELATED WORK

A. Lap Time Prediction Studies

Recent research has increasingly focused on predicting lap times in Formula 1 as a key input for strategic decision-making. In [1], an LSTM-based model was trained on historical data from 2014 to 2023 to predict lap times and driver positions using features such as grid position, pit stop status, tire compound, and time gaps. Similarly, [5] employed deep learning techniques to estimate lap times during races, showing that LSTM-based architectures outperform classical regression models in sequential prediction tasks.

A more strategy-oriented approach is presented in [3], where driver archetypes and safety car scenarios are analyzed for their impact on lap times and pit strategies. These studies collectively highlight the increasing relevance of time-series modeling—particularly LSTM networks—in capturing the nonlinear and dynamic nature of Formula 1 race data.

B. Race Strategy and Pit Modeling

Beyond lap time prediction, various studies have addressed the role of pit stop strategy and race conditions in shaping driver performance and overall race outcomes. In [7], a predictive model is proposed to assess how pit stop timing, clean air exposure, and yellow flag conditions influence race positioning. The study uses real-world F1 data to model race outcomes under various strategy scenarios.

A reinforcement learning–based simulation of race decision-making is presented in [9], where pit stops, tire wear, and fuel load are dynamically optimized. These approaches emphasize the complexity of race strategy modeling and the necessity of incorporating multiple interacting variables into AI-based decision-support systems.

C. Deep Learning Methods for Time Series

A wide range of deep learning architectures have been explored for time series classification and prediction, many of which are relevant to the modeling of sequential race data in Formula 1. In [13], a hybrid architecture combining Long Short-Term Memory (LSTM) and Fully Convolutional Networks (FCN) was proposed, demonstrating state-of-the-art performance across benchmark time series datasets. A multivariate extension of this approach is introduced in [14], where attention mechanisms and squeeze-and-excitation blocks further enhance model accuracy and representation learning.

Additionally, [12] proposes the Hidden-Unit Logistic Model (HULM) as a generative method for multivariate time series classification, providing an alternative to recurrent neural networks. These models, though not developed specifically for motorsports, offer valuable architectural insights that inform the selection and design of the LSTM-based model used in this study.

III. METHODOLOGY

Figure 1 illustrates the overall architecture of the proposed lap time prediction system. Raw inputs such as lap number, tire compound, pilot data, and race conditions are first preprocessed through normalization and encoding techniques. Feature engineering adds derived metrics like tire age and pilot rating, which are then fed into an LSTM network to predict lap times.

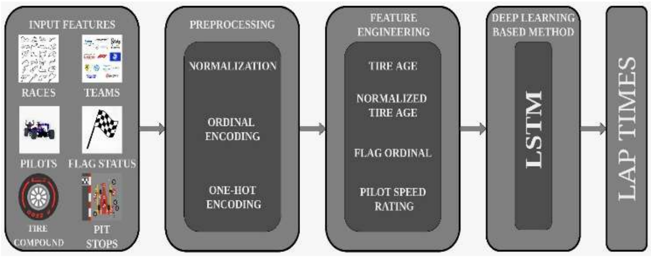
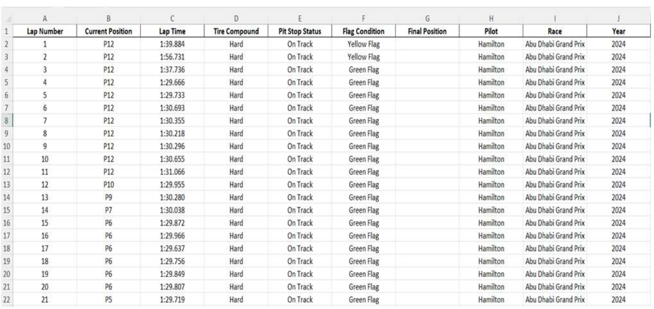


Fig. 1. Block diagram illustrating the proposed methodology for lap time prediction in Formula 1 using deep learning-based modeling

A. Dataset

The dataset used in this study consists of real Formula 1 race telemetry and lap-by-lap performance data collected from the 2021 to 2024 seasons. The data was compiled from publicly available race result platforms, parsed using automated web scraping tools, and structured into a time-series format suitable for model training. Each data instance corresponds to a single completed lap and includes multiple features such as lap time, tire compound, tire age, pit stop status, race flag condition, track status, driver identity, team, and race identifier. The final dataset spans four complete seasons, comprising 75 races, 21 drivers, and more than 80,000 laps and 13 base features, offering a diverse representation of track types, driver behaviors, and race conditions. As a final example, a representative portion of the dataset containing lap-by-lap telemetry from the 2024 Formula 1 races is shown in Fig. 2.



Lap Number	Current Position	Lap Time	Tire Compound	Pit Stop Status	Flag Condition	Final Position	Pilot	Race	Year
1	P12	1:39.884	Hard	On Track	Yellow Flag		Hamilton	Abu Dhabi Grand Prix	2024
2	P12	1:56.731	Hard	On Track	Yellow Flag		Hamilton	Abu Dhabi Grand Prix	2024
3	P12	1:37.736	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
4	P12	1:29.666	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
5	P12	1:29.733	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
6	P12	1:30.693	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
7	P12	1:30.355	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
8	P12	1:30.218	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
9	P12	1:30.296	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
10	P12	1:30.655	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
11	P12	1:31.066	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
12	P10	1:29.955	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
13	P9	1:30.280	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
14	P7	1:30.038	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
15	P6	1:29.872	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
16	P6	1:29.966	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
17	P6	1:29.637	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
18	P6	1:29.756	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
19	P6	1:29.849	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
20	P6	1:29.807	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024
21	P5	1:29.719	Hard	On Track	Green Flag		Hamilton	Abu Dhabi Grand Prix	2024

Fig. 2. A representative snapshot from the dataset including lap by lap data from the 2024 Formula 1 races

Feature engineering was applied to normalize tire age across compound types and to encode categorical variables such as driver and team IDs. The resulting dataset allows for temporal learning while preserving key contextual information for accurate lap time prediction.

B. LSTM Model

To model the sequential and temporal dependencies inherent in lap-by-lap race data, a Long Short-Term Memory (LSTM) architecture was adopted. The model is designed to capture both short-term and long-term patterns in driver performance across laps. The input to the model is a fixed-length window of previous laps, each represented by a multivariate feature vector. The model architecture consists of stacked LSTM layers followed by fully connected dense layers, which map the temporal output into a single continuous lap time prediction.

The base architecture includes three LSTM layers with 128, 64, and 32 units, respectively. These are followed by two dense layers with 128 and 64 units, and a final output node representing the predicted lap time. Dropout regularization is applied after each LSTM layer to reduce overfitting.

The model was implemented using the TensorFlow framework and trained using the Adam optimizer. Mean Squared Error (MSE) was used as the loss function due to the continuous nature of the prediction task. The detailed layer-wise architecture of the LSTM model used for lap time prediction is presented in Table I.

TABLE I. LSTM model architecture

Layer	Nodes	Other Operations
LSTM	128	Dropout (0.2), Recurrent Dropout (0.2)
LSTM	64	Dropout (0.2), Recurrent Dropout (0.2)
LSTM	32	Dropout (0.2)
Fully Connected (FCN)	128	Dropout (0.2), ReLU
Fully Connected (FCN)	64	ReLU
Fully Connected (FCN)	1	-

C. Model Training

The model was trained using a custom time-series dataset constructed from race telemetry between 2021 and 2023. The target variable was the lap time for the next lap, and each training instance consisted of a fixed-length sequence of past laps represented by multivariate feature vectors. Each input sequence was extracted using a fixed-length sliding window, eliminating the need for additional padding. The input sequence includes the lap times of the preceding laps (up to t), while the target is only the lap time at t+1. Thus, the loss is computed solely on the prediction of lap t+1, ensuring no future information leaks into the training Input features were scaled using a manual min-max normalization approach to preserve interpretability in the model outputs. This allowed for direct comparison between predicted and actual lap times in seconds, as opposed to scaled units. The model was trained using the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and a total of 200 epochs.

The training process utilized the Mean Squared Error (MSE) loss function, and performance was monitored both globally and per team-driver combination. To enhance generalization, dropout layers were incorporated after each LSTM layer.

During the training phase, several architectural and hyperparameter refinements were applied. The number of LSTM layers was increased from one to three to improve the model's capacity to represent complex race patterns. The sequence window was also expanded to capture longer-term temporal dependencies. These changes resulted in improved alignment between predicted and actual lap time trends. The training parameters used for the LSTM model are summarized in Table II.

TABLE II. Training parameters

Parameter	LSTM Value
Learning Rate	0.001
Epoch	200
Batch	32
Optimization Algorithm	ADAM

IV. EXPERIMENTAL RESULTS

The model was evaluated under two distinct configurations: one using the full dataset containing all laps, and another using a selectively filtered dataset in which laps influenced by red flag periods, unexpected restarts, extreme weather conditions, and safety car interventions were excluded. The evaluation presented below is based exclusively on this filtered dataset, as it better represents clean racing conditions and allows for a more accurate assessment of the model's predictive performance.

A closer inspection shows that laps longer than ~110 seconds are predicted with lower accuracy. These cases typically correspond to pit-in/pit-out events, safety car periods, or extreme weather conditions. Although such laps were largely excluded in the filtered dataset to approximate clean racing conditions, some of them remain, and their underrepresentation makes them harder to model accurately. As a result, the predictor remains well-calibrated under normal racing conditions but faces difficulties with these rare and irregular events. While these outlier laps may produce larger deviations, their practical impact on strategy is limited, since such laps are usually treated separately by teams and are not central to pit stop or pace optimization.

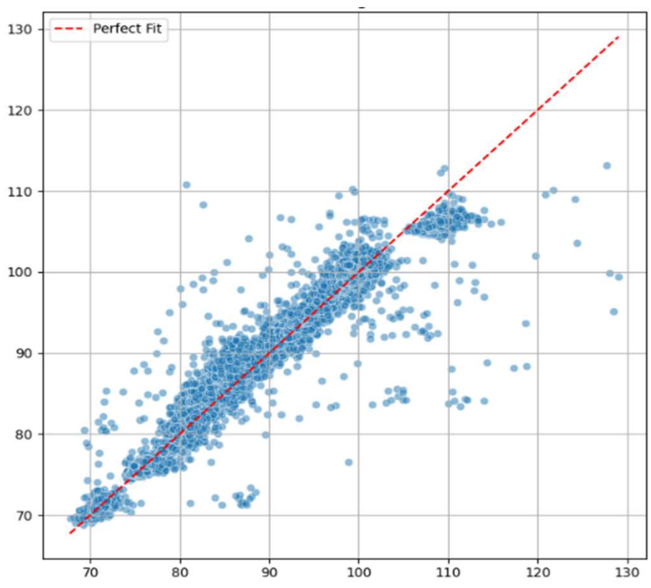


Fig. 3. Scatter plot of predicted vs. actual lap times on clean laps

As shown in Figure 3, the scatter plot between predicted and actual lap times reveals a strong linear correlation. The model achieved an R^2 score of 0.950, with a Mean Absolute Error (MAE) of 1.404 seconds and a Mean Squared Error (MSE) of 5.236. This indicates that the model can reliably track lap time trends and deliver accurate estimates under stable race conditions.

The residual distribution shown in Figure 4 demonstrates that the majority of prediction errors lie within ±5 seconds of the actual lap times, forming a near-symmetric and centered histogram. This suggests that the model does not exhibit systemic bias and is well-calibrated for normal racing conditions.

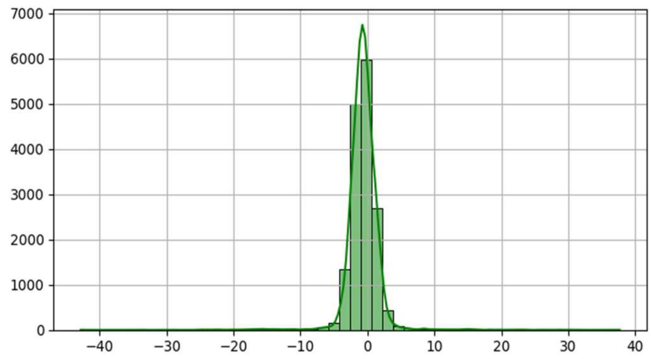


Fig. 4. Error distribution of lap times

To further illustrate the model's behavior on a driver level, Figure 5 presents lap-by-lap predictions versus actual times for Charles Leclerc during the Abu Dhabi Grand Prix. The model successfully follows the general performance trend and adapts to mid-race fluctuations with minimal deviation, confirming its ability to preserve temporal consistency across multiple laps.

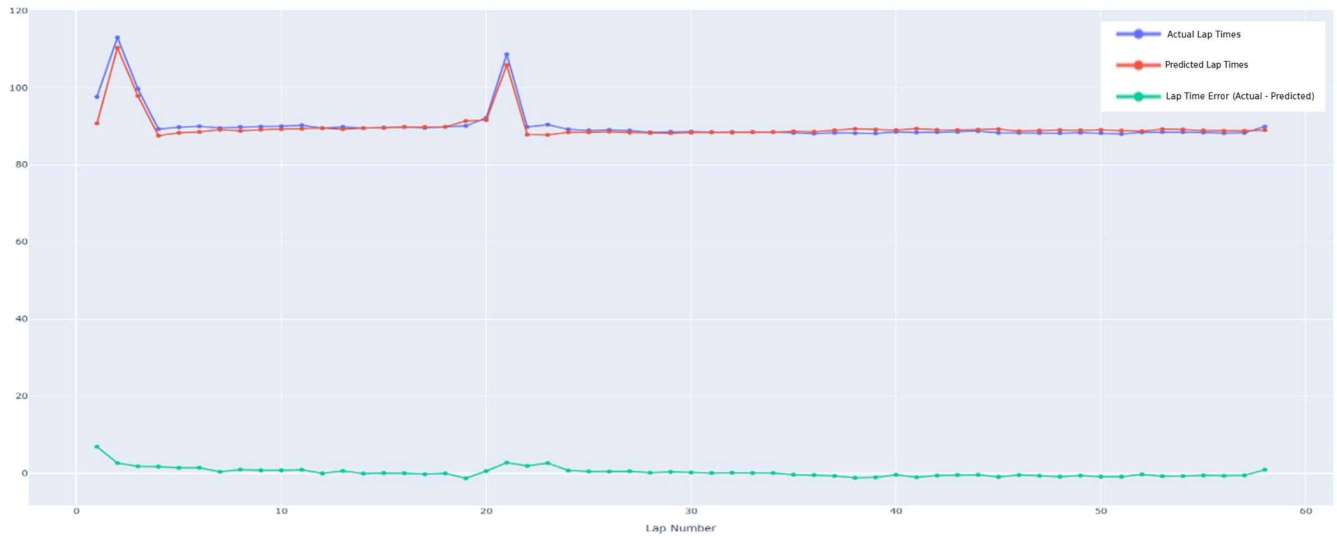


Fig. 5. Lap-by-lap predicted vs. actual lap times for Charles Leclerc in the Abu Dhabi GP.

V. CONCLUSION

This study focused on forecasting future lap times in Formula 1 by leveraging historical race data and implementing an LSTM-based time series model. A comprehensive dataset spanning four seasons (2021–2024) was constructed, incorporating diverse variables such as track conditions, tire compounds, and driver performance characteristics.

Following an extensive data preprocessing and feature engineering pipeline, the model was trained to learn sequential dynamics across laps using a deep LSTM architecture. The resulting predictions were evaluated using both quantitative metrics and visual analysis. In particular, tests on the 2024 season demonstrated the model's ability to generate accurate and stable lap time forecasts.

One of the model's key strengths lies in its ability to not only perform retrospective predictions but also serve as a foundation for in-race decision support. Lap time estimates generated by the model can assist strategic decision-making related to pit stops, tire management, and race pacing. For example, in an undercut situation—where a driver pits earlier than a rival to gain the advantage of fresher tires—the ability to forecast whether the first laps after the stop will be significantly faster is critical to regaining track position. Conversely, in an overcut scenario—where a driver stays out longer while a competitor pits—the prediction of lap times on worn tires for one or two additional laps determines whether it is possible to maintain the lead before eventually stopping. While the present model is not yet sufficient for direct deployment, these cases illustrate how lap-time forecasting can form a quantitative layer that supports, rather than replaces, strategic choices during a race.

A significant outcome of the study is the ability to perform team-level and driver-level evaluations. The model exhibited distinct behavior across different teams, reflecting tactical variations and performance patterns that differ by car characteristics and circuit types. While these patterns were not explicitly hardcoded, the model was able to implicitly capture them through its predictive behavior. Additionally, driver-specific prediction accuracy suggests the model is capable of learning individual driving styles and tendencies. This opens up the possibility of building personalized strategy assistants based on driver-specific profiles.

The results indicate that this model can form a solid foundation for future AI-assisted racing applications. Future enhancements may include integrating real-time telemetry, modeling DRS, SC, and VSC conditions, and comparing LSTM against alternative architectures. Furthermore, incorporating this predictive model into a real-time decision support system could enable intelligent recommendations, such as optimal pit window detection or tire selection advisories.

In conclusion, this research demonstrates that it is feasible to develop an AI-driven analytical assistant for high-stakes environments such as Formula 1. The LSTM model delivers meaningful performance predictions and offers promising potential for integration into real-time race strategy systems—making it a valuable contribution to both academic and applied motorsport analytics.

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